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Atomic-Layer-Deposited ZnF_2 on Aluminum Enabling *in situ* $\text{NaZn}_{13}/\text{NaF}$ Interphase for Robust Anode-Free Sodium Batteries

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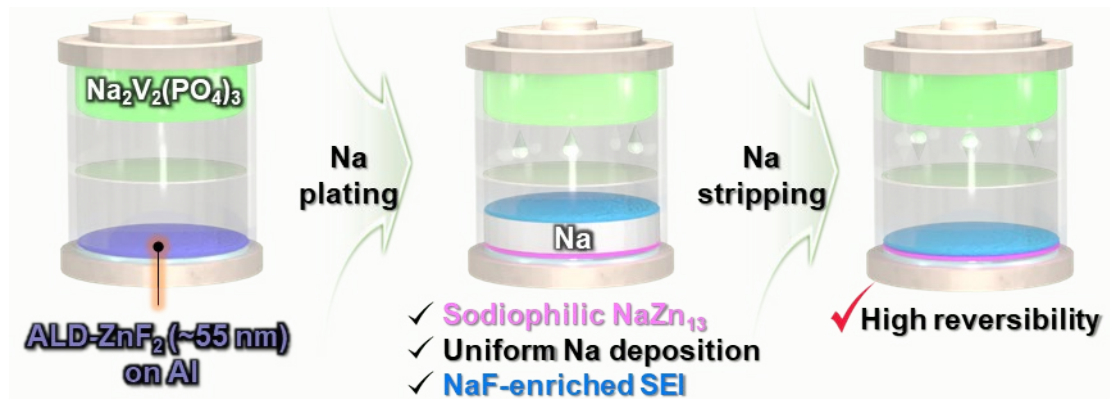
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“Sodiation of an ultrathin ALD-ZnF₂ layer on an Al current collector generates a sodiophilic NaZn₁₃ alloy and a NaF-rich SEI, enabling uniform Na nucleation and highly reversible anode-free sodium batteries.”



Abstract

“Anode-free” sodium batteries offer high energy density but suffer from rapid capacity decay due to the poor reversibility and non-uniformity of sodium plating/stripping on the anode-side current collector. In this study, we demonstrate a novel and efficient current collector consisting of aluminum foil coated with an ultrathin ZnF_2 layer to address these challenges. Upon initial electrochemical sodiation, the ZnF_2 layer is converted *in situ* into a NaZn_{13} alloy and NaF. The NaZn_{13} alloy is expected to enhance sodiophilicity, markedly reducing the nucleation overpotential and promoting uniform Na nucleation, while the NaF component strengthens the solid-electrolyte interphase, ensuring the Na^+ flux and stabilizing Na deposition. Owing to this synergistic NaZn_{13} /NaF interphase, the ZnF_2 -modified Al current collector delivers a high Coulombic efficiency of $\sim 99.7\%$ by a depositing/stripping capacity of 1 mAh cm^{-2} under a current density of 0.5 mA cm^{-2} . Furthermore, anode-free Na cells that pair this collector with $\text{Na}_3\text{V}_2(\text{PO}_4)_3$ cathodes (mass loading: 11.5 mg cm^{-2}) exhibit markedly prolonged cycle life. This work establishes, to the best of our knowledge, the first example of an ALD-derived ZnF_2 coating on Al that is deliberately converted into a NaZn_{13} /NaF interphase for anode-free sodium batteries, offering a simple and scalable interfacial-engineering strategy for safe, high-performance sodium-based energy storage systems.

Keywords: anode-free sodium battery, sodiophilic, NaF-rich SEI, cycling stability, dendrite

Introduction

Anode-free sodium batteries (AFSBs) have emerged as a promising architecture for next-generation sodium (Na) storage because they **theoretically promise** higher energy density, simplified manufacturing, improved process safety, and reduced production costs compared with conventional Na batteries.^{1,2} In this configuration, no **anode**-active material is present initially; instead, **Na originating from the cathode is directly plated** onto the current collector (CC) at the anode side during the first charge.^{2,3} However, the plated Na often exhibits porous or mossy morphologies, which promote the formation of electrochemically inactive Na and irreversible Na⁺ consumption via side reactions.^{4,5} **Without supplemental** metallic Na to compensate for these losses, the capacity of AFSBs rapidly fades.⁶

The behavior of Na deposition in such systems is mainly governed by two factors: the sodiophilicity of the CC and the stability of the solid–electrolyte interphase (SEI). Aluminum (Al) foil, the most common anode-side CC, is intrinsically sodiophobic and requires a high nucleation overpotential, leading to **inhomogeneous** nucleation, locally amplified current densities, and dendritic Na growth.^{7–9} Although modifying Al with sodiophilic components (Au, Ag, Zn, Sn, etc.) can lower the nucleation barrier,^{10–13} the resulting SEI often remains fragile, so **that** cracks formed during repeated plating/stripping expose fresh Na and further accelerate dendrite formation.^{14–17}

SEI engineering, therefore, plays a crucial role in stabilizing Na metal. Among various strategies, enriching the SEI with sodium fluoride (NaF) has proven particularly effective because NaF combines a low Na⁺ diffusion barrier with excellent mechanical and chemical stability and a high interfacial energy toward Na.^{18,19} NaF-rich, inorganic SEI layers can suppress parasitic reactions and efficiently mitigate dendritic growth, significantly extending the lifetime of Na metal anodes.^{20–22}

Recent work has shown that integrating Na-alloying metals with fluorinated species is especially beneficial **for homogenous Na plating and stripping**. A representative example is the study by Wang et al., who screened metal fluorides coated on Al (MF_x@Al) and identified

BiF₃-derived interphases - forming Na₃Bi and NaF *in situ* - as particularly effective, delivering outstanding Coulombic efficiency and high-rate capability at both ambient and subzero temperatures.²³ This work clarified the synergistic roles of Na-alloying phases and NaF-rich SEI and introduced Na adsorption energy together with a lattice mismatch as quantitative design criteria. However, such approaches rely predominantly on slurry-cast metal-fluoride layers, and there remains a lack of interphases in which an ultrathin metal-fluoride film is grown with atomic-scale precision and then deliberately converted into a Na-alloy phase and NaF for AFSBs.

Here, we address this gap by, to the best of our knowledge, demonstrating the first ALD-derived ZnF₂ coating on Al that *in situ* forms a NaZn₁₃/NaF interphase tailored for anode-free sodium batteries. Atomic layer deposition (ALD) affords Å-level thickness control and excellent conformality.^{24–26} enabling the formation of a continuous ZnF₂ layer over the entire Al surface and a nearly simultaneous, spatially homogeneous phase transformation during the initial sodiation step. We show that this conformal ZnF₂ layer is converted into a NaF/NaZn₁₃ interphase, in which a highly sodiophilic NaZn₁₃ phase serves as a uniform nucleation template, and NaF constructs a robust and stable SEI that effectively regulates Na⁺ flux. Using cross-sectional TEM, XPS depth profiling, and DFT calculations, we provide direct structural and chemical evidence for this transformation and quantitatively correlate the interphase architecture with Na nucleation and full-cell performance. As a result, the ZnF₂-modified Al current collector enables dense, dendrite-free Na deposition, significantly reduced nucleation overpotential, and highly reversible Na plating/stripping, leading to improved rate capability, prolonged cycle life, and increased energy density in anode-free Na cells with Na₃V₂(PO₄)₃ (NVP) cathodes.

Results and Discussion

Design Rationale for the ALD ZnF₂ Interphase

We recognized that the inherent sodiophobicity of bare Al foil, along with the formation of an uneven SEI on its surface, poses significant challenges to achieving uniform Na deposition.²⁷ These unfavorable interfacial characteristics lead to poor reversibility of Na plating/stripping and thus limit the direct use of bare Al foil as a current collector (CC) in AFSBs (Figure S1). To address this issue, we hypothesized that coating Al foil with a suitable interfacial layer could markedly improve the uniformity of Na plating/stripping.

Guided by the Na–Zn phase diagram (Figure 1A),²⁸ we determined that alloying between Na and Zn would yield the NaZn₁₃ phase, which is thermodynamically stable at room temperature. Density functional theory (DFT) calculations revealed that NaZn₁₃ exhibits negative adsorption energies (Figure 1B and Figure S2), indicating strong binding with Na and thus high sodiophilicity. Therefore, the NaZn₁₃ alloy is expected to lower the Na nucleation barrier and promote uniform Na deposition.

An ideal coating material should satisfy two key requirements: (i) it should be highly sodiophilic to facilitate Na nucleation, and (ii) it should promote the formation of a NaF-rich SEI to guide uniform Na growth. Based on these criteria, we focused on ZnF₂, which combines sodiophilic Zn with F, a known promoter of NaF formation within the SEI. In addition, ZnF₂ itself exhibits strong sodiophilicity, as evidenced by its highly negative adsorption energies (Figure 1C). We therefore hypothesized that electrochemical reactions would *in situ* convert ZnF₂ into NaZn₁₃ and NaF. This process would simultaneously introduce a sodiophilic Na-alloying phase (NaZn₁₃) and a NaF-rich SEI, both of which are expected to enhance Na nucleation and growth behavior (Figure 1D). To deposit the ZnF₂ layer onto Al foil, we employed ALD, a well-established technique in interfacial engineering that offers precise thickness control, excellent uniformity, and conformal coverage, particularly suited for energy storage applications.^{24–26,29,30} The experimental results presented below strongly support this design hypothesis.

Structural and Chemical Characterization of ALD-ZnF₂ on Al

At the initial stage of our study, ZnF₂/Al samples with varying ZnF₂ thicknesses were fabricated by controlling the number of ALD cycles. After comprehensive characterization, the sample produced using 625 ALD cycles was identified as having the most suitable ZnF₂ thickness and was selected for subsequent analyses. As shown in the optical images in Figures 2A and 2B, the Al foil became noticeably less reflective after ALD treatment, indicating a clear change in surface appearance. Corresponding scanning electron microscope (SEM) images revealed a more uniform surface texture and smoother morphology on the ALD-treated Al foil.

The coating thickness was determined to be ≈55 nm from cross-sectional transmission electron microscope (TEM) (Figure 2C). Energy-dispersive X-ray spectroscopy (EDX) mapping confirmed a homogeneous distribution of Zn and F across the Al surface (Figures 2D–F and Figure S3), demonstrating that the ALD process produced a conformal and compositionally uniform ZnF₂ layer.

To further elucidate the structure and composition of the ZnF₂ layer, we employed X-ray diffraction (XRD) and X-ray photoelectron spectroscopy (XPS). Distinct diffraction peaks corresponding to ZnF₂ in the XRD patterns provided clear evidence of ZnF₂ formation on the Al substrate (Figure 2G).^{26,31} Notably, the XPS survey spectrum of ZnF₂/Al did not show any characteristic Al peaks, indicating that the ZnF₂ layer was sufficiently thick to fully cover the substrate (Figure S4). In addition, the measured doublet separation energy between the Zn 2p_{3/2} (1022.5 eV) and Zn 2p_{1/2} (1045.6 eV) peaks was 23.1 eV (Table S1), in agreement with the typical value for Zn²⁺ species (Figure 2H).^{32–34} The F 1s spectrum exhibited a main peak at 685.8 eV, attributable to Zn–F bonding (Figure 2I). Collectively, these results confirm the successful deposition of a continuous, conformal, and ultrathin ZnF₂ coating on Al by ALD.

Initial Electrochemical Activation and *in situ* ZnF₂ Conversion

To assess the role of ZnF₂ in modulating Na plating/stripping behavior, we assembled coin cells using either ZnF₂/Al or bare Al as the working electrode, paired with Na foil (denoted Na||ZnF₂/Al and Na||Al, respectively). The cells were first discharged to 0.01 V vs. Na⁺/Na and

then cycled five times between 0.01 and 1.0 V vs. Na⁺/Na to stabilize the SEI (Figure S5). Notably, while the Na||Al cell exhibited a rapid voltage drop, the Na|| ZnF₂/Al cell showed a pronounced voltage plateau at around 0.3 V during the initial discharge (Figure 3A). This plateau is attributed to the electrochemical reactions between ZnF₂ and Na⁺, leading to the formation of NaZn₁₃ alloy and NaF according to:



To gain further insight into the interfacial changes associated with this activation step, we disassembled the cells for post-mortem characterization. Remarkably, the bare Al electrode retained its metallic luster, whereas the ZnF₂/Al electrode turned uniformly black (Figures 3B and 3C), indicating a substantial change in surface chemistry. SEM images of bare Al revealed only minor surface modifications, with a few scattered deposits likely arising from electrolyte decomposition. In contrast, the ZnF₂/Al surface was completely and uniformly covered with reaction products generated during sodiation of the ZnF₂ layer. The resulting layer appeared expanded relative to the pristine ZnF₂ layer.

Evidence for NaF/NaZn₁₃ Interphase Formation

The structural evolution of the ALD-ZnF₂ layer upon sodiation was further investigated using cryo-STEM combined with electron energy-loss spectroscopy (EELS).^{35–37} The cross-sectional annular dark field scanning transmission electron microscopy (ADF-STEM) image of the sodiated ZnF₂/Al (Figure 3D), acquired simultaneously during EELS spectrum imaging, reveals multiple distinct layers above the Al substrate. Note that the ADF-STEM image was recorded under low-angle collection conditions during the EELS acquisition; therefore, the image contrast arises from a mixture of diffraction and mass-thickness scattering rather than Z-contrast, and should not be directly interpreted as compositional contrast.³⁸ The overall

thickness of the reacted zone is significantly larger than that of the as-deposited ZnF_2 film (~ 55 nm), consistent with volume expansion during Na insertion and the conversion reaction.

To identify the compositional origin of each layer, EELS line profiles were acquired across the reaction zone from point I to point II (Figure 3E). The characteristic Zn L-edge and Na K-edge signals reveal a clear spatial separation of elements within the reacted layer. The Zn signal is concentrated near the Al substrate, while the Na signal is strong throughout the reaction layer and extends toward the outer surface. Representative EELS spectra extracted from each region, including the F K-edge (~ 685 eV), Zn L-edge (~ 1020 eV), and Na K-edge (~ 1072 eV), are presented in Figure S6.³⁹ These spectra suggest a layered elemental distribution consistent with the following phase sequence, as corroborated by XPS analysis (Figure 3H and 3I), subsequently: unreacted ZnF_2 at the bottom, a partially converted $\text{ZnF}_2/\text{NaF}/\text{Zn}$ mixed zone, a NaF-dominant middle layer, and a NaZn_{13} layer at the top. This spatial arrangement is consistent with a reaction pathway in which ZnF_2 is progressively reduced as Na is incorporated from the electrolyte side inward.

XPS measurements provided complementary chemical-state information that corroborates the EELS spatial analysis. The F 1s spectrum of the sodiated ZnF_2/Al electrode showed that the F–Zn signal became extremely weak, indicating that most ZnF_2 had been consumed (Figure 3H and Figure S7A). In contrast, the F–Na signal at 684.7 eV⁴⁰ was significantly intensified compared with bare Al, confirming substantial *in situ* formation of NaF from ZnF_2 . The high intensity of the NaF signal in the surface region indicates that the SEI formed on ZnF_2/Al is NaF-rich, which is expected to enhance mechanical robustness and ion transport.

The Zn 2p spectrum of the sodiated ZnF_2/Al electrode was dominated by peaks corresponding to metallic Zn^0 ,⁴¹ confirming electrochemical reduction of Zn^{2+} in ZnF_2 (Figure 3I 3H and Figure S7B). The Na 1s region exhibited two distinct components: one associated with metallic Na^0 in the NaZn_{13} alloy and another corresponding to ionic Na^+ in NaF. Collectively, these results confirm the *in situ* transformation of ZnF_2 into NaZn_{13} alloy and NaF,

both of which are expected to play key roles in regulating Na nucleation and ensuring uniform Na deposition.

Impact of NaZn₁₃/NaF Interphase on Na Nucleation and Growth

The effectiveness of the *in situ*-generated NaZn₁₃ alloy and NaF in guiding Na nucleation and growth was evaluated by Na electrodeposition tests at different current densities and cut-off capacities. At a current density of 0.5 mA cm⁻², the nucleation overpotential for Na on ZnF₂/Al was only 17 mV, which is an order of magnitude lower than the 173 mV observed for bare Al (Figure 4A). During stripping, the Na||Al cell exhibited irregular voltage fluctuations (highlighted in Figure 4A), indicative of non-uniform and unstable Na dissolution. In contrast, the Na||ZnF₂/Al cell shows smooth stripping profiles. Even at a higher current density of 2 mA cm⁻², ZnF₂/Al maintained a markedly lower overpotential than bare Al (Figure S8), reflecting the  improved sodiophilicity of the NaZn₁₃-containing interphase.

Morphological analysis further highlights the beneficial effect of the NaZn₁₃/NaF interphase. SEM images taken at an early deposition stage showed that Na nucleates as isolated clusters on bare Al (Figure 4B). This uneven distribution, amplified by “the tip effect”, drives subsequent Na⁺ to deposit preferentially at existing protrusions, ultimately yielding a cracked, porous layer riddled with voids. After deposition to 2 mAh cm⁻², the Na layer on bare Al exhibits significant thickness variations and exceeded ≈26 μm (Figure S9A). Even after full Na stripping, residual Na clusters remained on the Al surface, underscoring the poor reversibility of Na plating/stripping and the propensity for active Na loss.

In sharp contrast, Na deposition on ZnF₂/Al followed a much more uniform pattern (Figure 4C). The finely dispersed NaZn₁₃ and residual ZnF₂ likely acted as sodiophilic nucleation sites, promoting film-like Na growth with minimal aggregation. At the same areal capacity of 2 mAh cm⁻², the Na layer on ZnF₂/Al was only ≈23.6 μm thick and displayed much less thickness variation (Figure S9B).  This suggests that the NaF-rich SEI could regulate Na⁺ flux, resulting in compact and uniform Na deposits. After complete stripping, the ZnF₂/Al surface

appeared clean and largely free of residual Na, indicating excellent reversibility of Na plating/stripping on this interphase.

Coulombic Efficiency and Long-Term Cycling Stability

Beyond morphological uniformity, the Coulombic efficiency (CE) of Na plating/stripping is a critical metric for **reversibility and thus predicting cycling stability**. To **quantify the CE**, a fixed amount of Na was repeatedly plated onto and stripped from bare Al and ZnF₂/Al CCs with a cut-off voltage of 1.0 V. At 0.5 mA cm⁻², the Al-based electrode could be cycled for up to ~100 cycles, but the CE fluctuated significantly and frequently dropped in individual cycles, **indicating unstable Na plating/stripping behavior**. (Figure 5A). In contrast, the ZnF₂/Al electrode exhibited **a high** and stable CE over 150 cycles, averaging 99.7%. At a high current density of 2 mA cm⁻², the CE of the Al electrode became highly erratic, whereas ZnF₂/Al maintained consistently high CE values even after 200 cycles, indicating more reversible Na plating/stripping.

The different CE trends are closely correlated with the Na deposition morphologies (Figure 5B). After the CE tests, Na plated on bare Al exhibits a rough, cracked, and highly porous structure. These surface cracks markedly enlarge the electrochemically active area, accelerate continuous side reactions with the electrolyte, and promote uncontrolled SEI thickening—all of which contribute **to Na** loss and rapid degradation. In contrast, Na deposited on ZnF₂/Al forms a smooth, dense, and nearly crack-free layer. This compact morphology suppresses parasitic reactions and minimizes active Na isolation, which is consistent with the high and stable CE observed for ZnF₂/Al.

To further assess long-term stability under more realistic conditions, symmetric cells were assembled using Na-plated Al or Na-plated ZnF₂/Al (pre-deposited Na capacity: 2 mAh cm⁻²) as both electrodes. The cells were cycled at 1 mA cm⁻² with a cut-off capacity of 1 mAh cm⁻². Symmetric cells with Na/Al electrodes failed after a relatively short time, showing steadily increasing voltage hysteresis **indicative of a growing** interfacial resistance and unstable Na

plating/stripping. In stark contrast, symmetric cells with Na/ZnF₂/Al electrodes operated stably for over 1000 h with a small and nearly constant hysteresis of ≈ 17 mV (Figure 5C). This outstanding durability confirms that the NaZn₁₃/NaF interphase generated from the ALD-ZnF₂ layer enables highly reversible Na cycling, making ZnF₂/Al a robust current-collector platform for both anode-less and anode-free sodium battery systems.

Full-Cell Performance in Lean-Na and Anode-Free Configurations

Finally, we evaluated the electrochemical performance of ZnF₂/Al CCs in both lean-Na and anode-free full-cell configurations with NVP cathodes. In the lean-Na configuration, a small amount of Na (0.5 mAh cm⁻²) was pre-deposited onto bare Al or ZnF₂/Al (denoted Na/Al and Na/ZnF₂/Al, respectively). The NVP cathode had a mass loading of 13.2 mg cm⁻², corresponding to a negative-to-positive capacity ratio of ≈ 0.33 . Both cells delivered an initial discharge capacity of 111 mAh g⁻¹ at 0.5 C (1 C = 117 mA g⁻¹; Figure 6A).

However, the cycling behavior of the two cells diverged significantly. The Na/Al || NVP cell exhibited rapid capacity decay to 75 mAh g⁻¹ within 100 cycles, accompanied by pronounced CE fluctuations. In contrast, the Na/ZnF₂/Al || NVP cell retained 101 mAh g⁻¹ after 100 cycles and 87 mAh g⁻¹ after 250 cycles, corresponding to capacity retentions of 91% and 78%, respectively. The average CE is close to 99.76% throughout cycling, indicating highly reversible Na plating/stripping. The Na/ZnF₂/Al || NVP cell also showed smoother charge–discharge profiles with lower polarization (Figure 6B), whereas the Na/Al || NVP cell exhibited abrupt voltage drops, likely associated with discontinuous Na stripping.

AFSB cells were then assembled using either bare Al or ZnF₂/Al CCs directly paired with NVP cathodes (mass loading: 11.6–12.2 mg cm⁻²). At a low rate of 0.1 C, the AFSB cells with bare Al suffered from rapid capacity fading and low CE within 30 cycles, indicating highly irreversible Na plating/stripping (Figure 6C). By contrast, the ZnF₂/Al-based AFSB cells showed much improved stability, retaining 55% of its initial capacity after 60 cycles with significantly higher CE under the same conditions. Even at 0.5 C, the ZnF₂/Al || NVP AFSB

cells maintained a capacity retention of 57% after 100 cycles (Figure 6D). In addition, the $\text{ZnF}_2/\text{Al} \parallel \text{NVP}$ cell exhibited smooth charge–discharge curves without visible voltage drops throughout cycling (Figures S10–S11). The ZnF_2/Al -based AFSB cells achieve a high energy density of 323 Wh kg^{-1} based on the total mass of active materials, surpassing those of several previously reported AFSBs in the literature (Figure 6E).

Taken together, these results demonstrate that the ALD-derived ZnF_2/Al current collector, through its *in situ* conversion to a $\text{NaZn}_{13}/\text{NaF}$ interphase, substantially enhances Na nucleation, suppresses dendrite formation, and improves reversibility of Na plating and stripping, leading to enhanced full-cell performance in both lean-Na and anode-free configurations.

Conclusions

In summary, we have demonstrated that an ultrathin ZnF_2 coating on Al foil is a highly effective strategy for engineering a high-performance current collector for stable AFSBs. During initial sodiation, the ZnF_2 layer is converted *in situ* into a conformal $\text{NaZn}_{13}/\text{NaF}$ interphase. The highly sodiophilic NaZn_{13} alloy markedly reduces the nucleation overpotential and promotes uniform Na deposition, while the NaF component stabilizes the SEI and likely regulates Na^+ flux to sustain even Na growth. The synergy between these two phases enables highly reversible plating/stripping, which beneficially impacts cycling stability and energy density. Therefore, both lean-Na and anode-free full cells exhibit markedly prolonged cycle life. To the best of our knowledge, this work provides the first deliberate realization and mechanistic elucidation of a $\text{NaZn}_{13}/\text{NaF}$ interphase generated from an ALD- ZnF_2 coating on Al for anode-free sodium batteries. Given its conceptual simplicity, scalability, and compatibility with foil-based electrode architectures, this ZnF_2/Al platform offers a practical and versatile route toward next-generation, high-performance sodium-based energy storage systems.

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Conflict of Interest

The authors declare no conflicts of interest.

Supporting Information

The Supporting Information is available at ...

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Figures

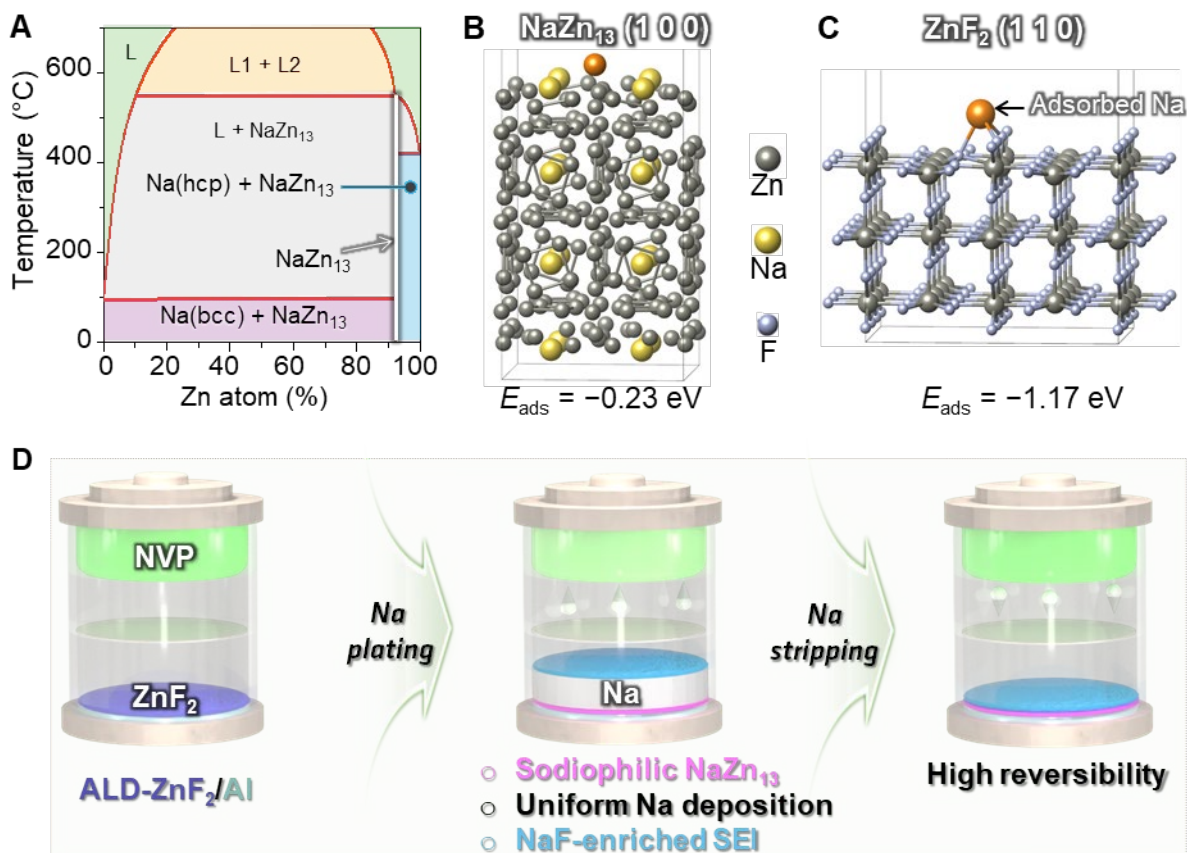


Figure 1. Schematic design of the ZnF₂/Al host for stable AFSBs. A) Na-Zn phase diagram.²⁸ B–C) DFT-optimized configurations and adsorption energies of Na on NaZn₁₃ (100) and ZnF₂ (110) surfaces, respectively. D) AFSB configuration using ZnF₂/Al as the current collector. An ultrathin ZnF₂ layer deposited by ALD is converted during the first Na plating (sodiation) into a NaF/NaZn₁₃ interphase; the sodiophilic NaZn₁₃ promotes uniform Na nucleation and growth, while the NaF-enriched SEI stabilizes the interface, enabling dense Na deposits and highly reversible cycling.

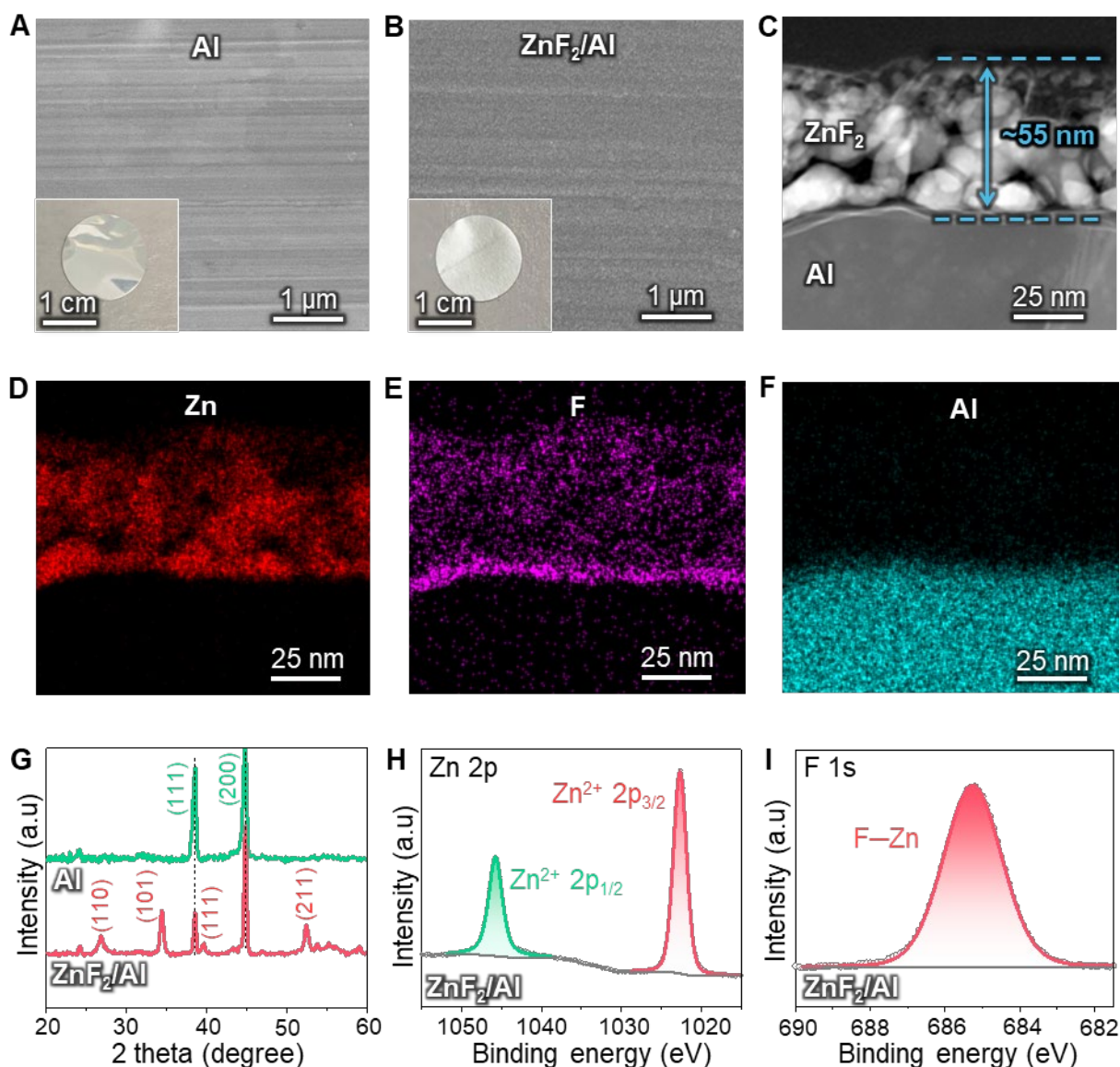


Figure 2. Structural and chemical characterization of the ALD-ZnF₂ coating on Al foil. **A–B)** SEM images of bare Al and ZnF₂/Al CCs, respectively (insets: corresponding optical photographs). **C)** Cross-sectional STEM-HAADF image showing an ultrathin ZnF₂ layer with a thickness of ~55 nm uniformly covering the Al substrate. **D–F)** STEM-EDX elemental maps of Zn, F, and Al, respectively. **G)** XRD patterns of bare Al and ZnF₂/Al. **H–I)** Fitted XPS spectra of Zn 2p and F 1s for ZnF₂/Al.

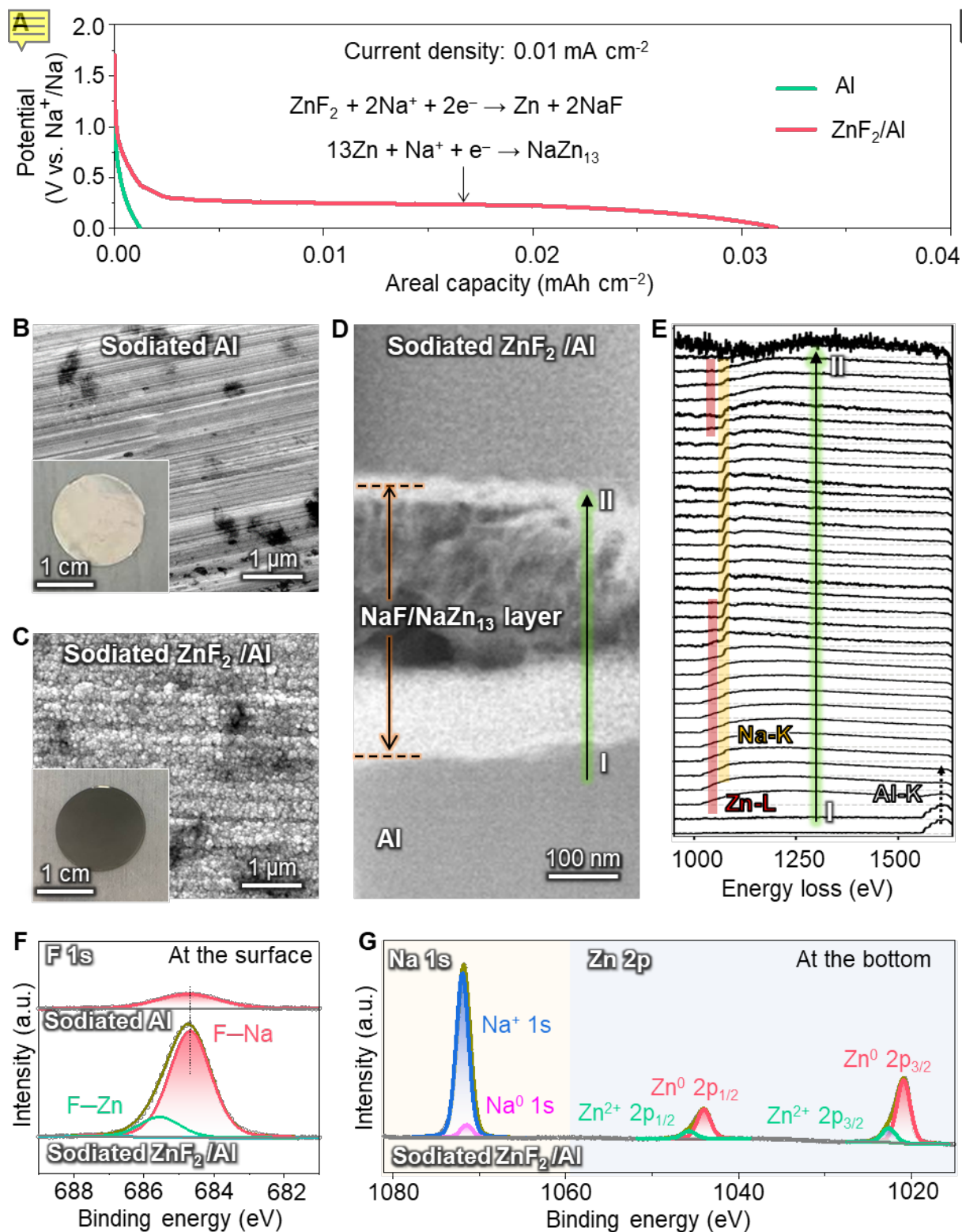


Figure 3. Electrochemical activation of the ALD-ZnF₂ layer and formation of the NaF/NaZn₁₃ interphase. **A)** First-discharge profiles of Na||Al and Na||ZnF₂/Al cells. **B–C)** SEM images of sodiated Al and sodiated ZnF₂/Al current collectors (insets: optical photographs). **D)** Cross-sectional TEM image of sodiated ZnF₂/Al. **E)** EELS line profiles acquired from point I to point II, showing the spatial distribution of the Zn L-edge, Na K-edge, and Al K-edge signals

across the reaction layer. **F** **(3)** Fitted XPS spectra of F 1s at the surface of the NaF/NaZn₁₃ layer **and** both Na 1s and Zn 2p at the bottom of the NaF/NaZn₁₃ layer **(3)**

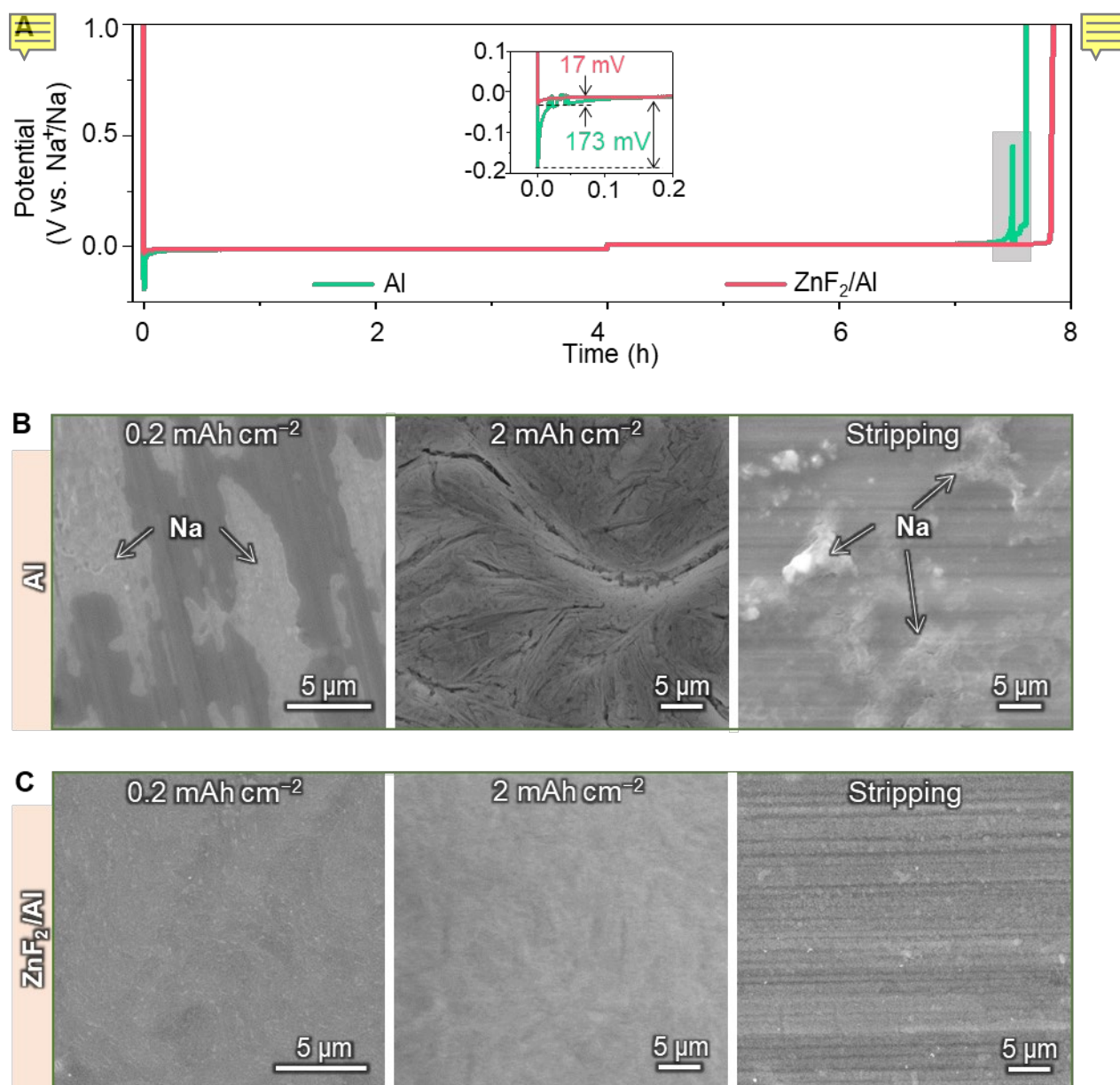


Figure 4. Effect of the $\text{NaZn}_{13}/\text{NaF}$ interphase on Na plating/stripping behavior.

A) Voltage profiles of Na plating/stripping on bare Al and ZnF_2/Al at 0.5 mA cm^{-2} . The inset shows the Na nucleation overpotentials. The grey region highlights irregular voltage fluctuations of the $\text{Na} \parallel \text{Al}$ cell. **B)** SEM images of Na deposited on bare Al at 0.2 and 2.0 mAh cm^{-2} , and after subsequent stripping. **C)** Corresponding SEM images for ZnF_2/Al .

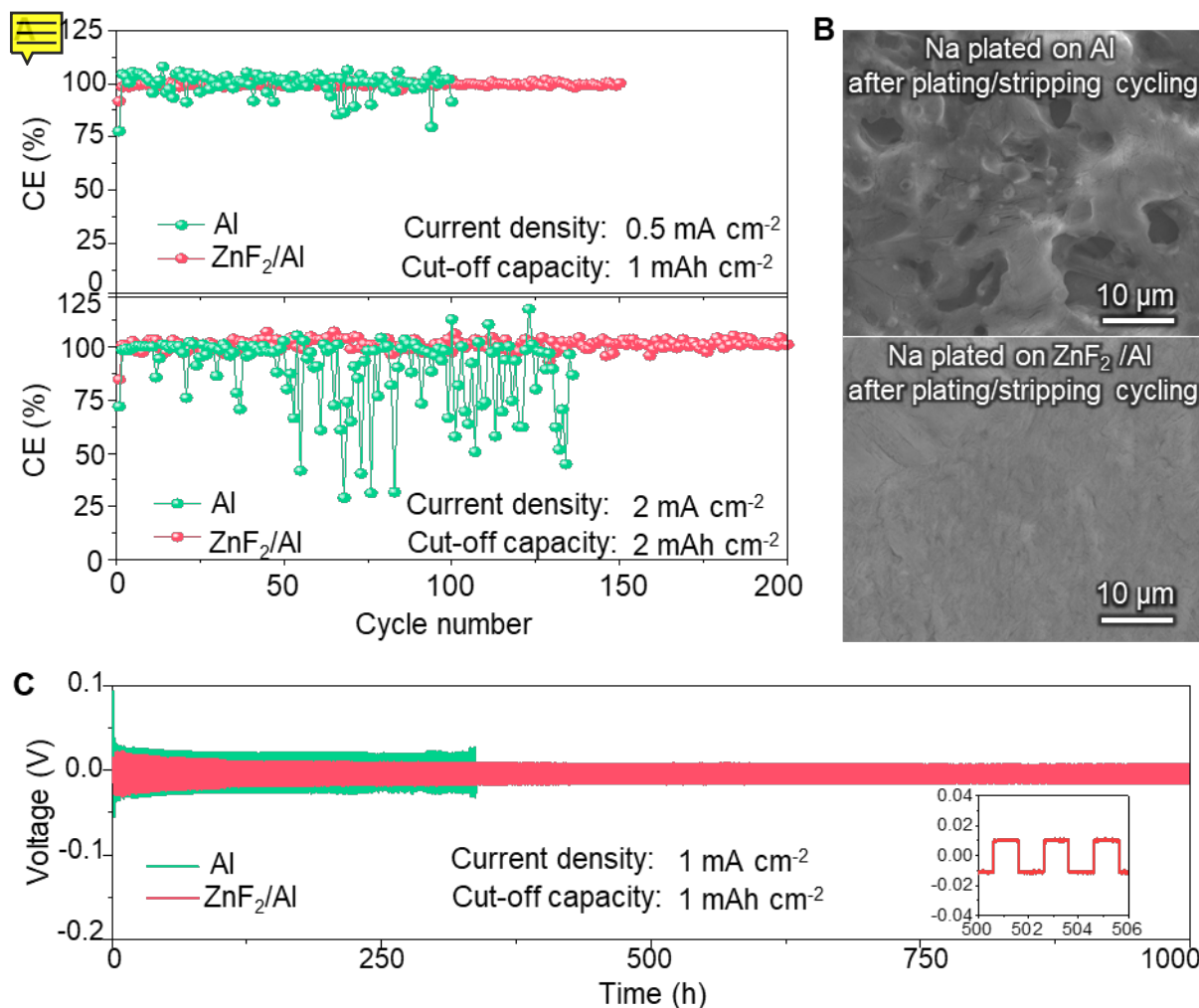


Figure 5. Coulombic efficiency, morphology, and long-term stability of Na plating/stripping. **A)** Coulombic efficiencies of Na plating/stripping on bare Al and ZnF₂/Al at current densities of 0.5 mA cm⁻² (cut-off capacity: 1 mAh cm⁻², top) and 2 mA cm⁻² (cut-off capacity: 2 mAh cm⁻², bottom). **B)** SEM images of Na deposits on Al (top) and ZnF₂/Al (bottom) after prolonged plating/stripping at 2 mA cm⁻². **C)** Galvanostatic cycling of symmetric Na/Al and Na/ZnF₂/Al cells at 1 mA cm⁻² with a cut-off capacity of 1 mAh cm⁻².

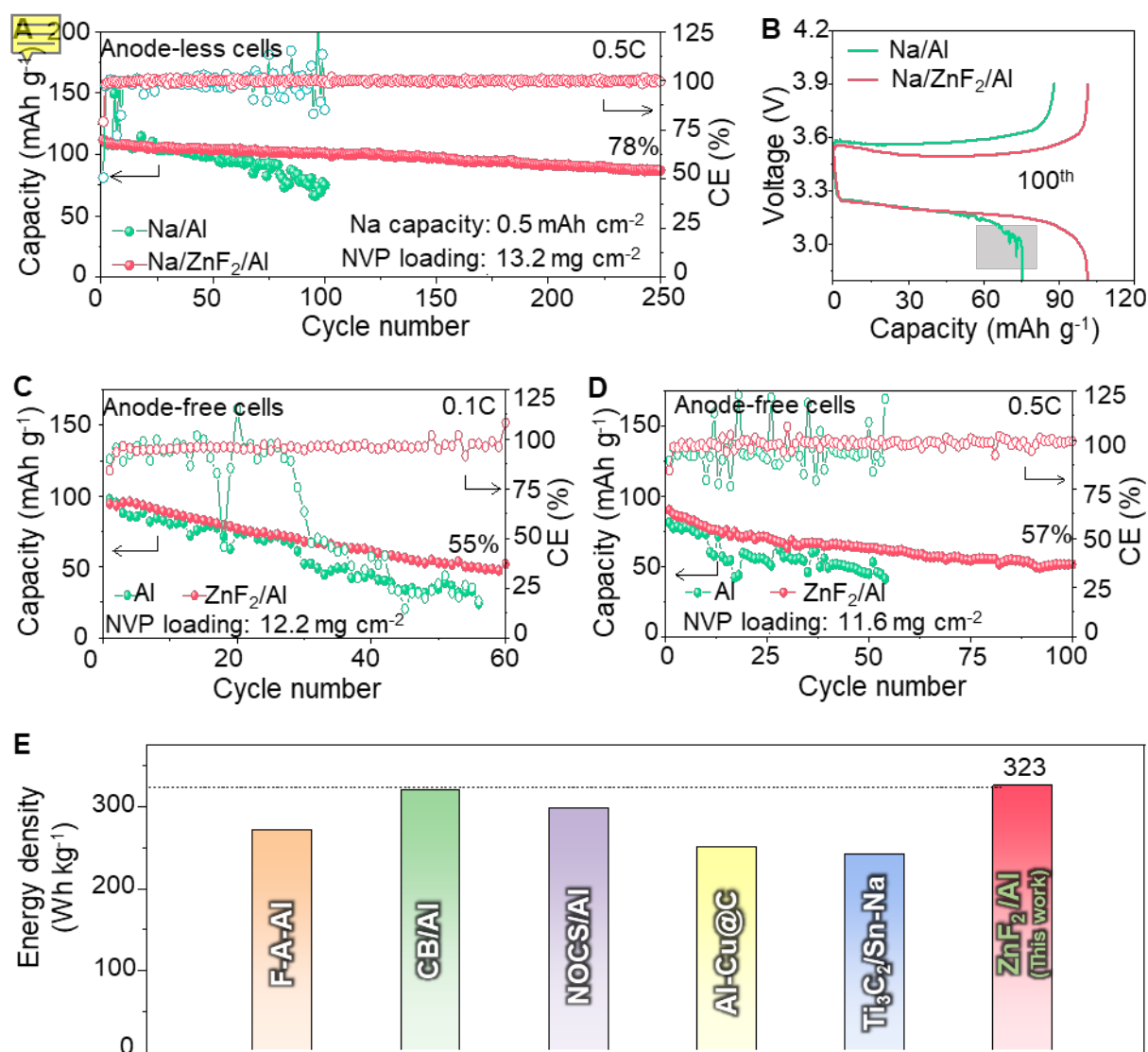


Figure 6. Full-cell performance of ZnF₂/Al current collectors in lean-Na and anode-free configurations. **A)** Cycling performance of Na/Al||NVP and Na/ZnF₂/Al||NVP lean-Na cells with a low N/P ratio at 0.5 C. **B)** Charge-discharge voltage profiles of Na/Al||NVP and Na/ZnF₂/Al||NVP cells at the 100th cycle. The grey region highlights irregular voltage fluctuations of the Na/Al||NVP cell. **C–D)** Cycling performance of Al||NVP and ZnF₂/Al||NVP anode-free cells at 0.1 C (C) and 0.5 C (D). **E)** Comparison of the energy densities of AFSBs calculated on the basis of the total mass of positive and negative electrode active materials, using different anode-side hosts: fluorinated Al (F-A-Al),⁹ carbon-black-coated Al (CB/Al),⁴² N,O-co-doped carbon-sphere-coated Al (NOCS/Al),⁴³ Cu-C-coated Al (Al-Cu@C),⁴⁴ MXene/Tin-Na (Ti₃C₂/Sn-Na),⁴⁵ and the present ZnF₂/Al.

SUPPORTING INFORMATION

Atomic-Layer-Deposited ZnF₂ on Aluminum Enabling *in situ* NaZn₁₃/NaF Interphase for Robust Anode-Free Sodium Batteries

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1. Experimental Section

Preparation of ZnF₂/Al

An ultrathin ZnF₂ layer was deposited onto Al foil by atomic layer deposition (ALD). Prior to deposition, commercial Al foil was cleaned with ethanol, dried, and then loaded into an ALD reactor (Savannah S200, Cambridge NanoTech Inc.). ZnF₂ was grown using diethylzinc (DEZ, Zn(C₂H₅)₂, Sigma-Aldrich) and an HF–pyridine solution (70 wt% HF, Sigma-Aldrich) as the Zn and F precursors, respectively. The ALD process was conducted in exposure mode under a N₂ carrier flow of 20 sccm at 150 °C and a pressure of ~0.1 Torr. Each ALD cycle consisted of a 0.01 s DEZ pulse, 20 s exposure, and 30 s purge, followed by a 0.5 s HF–pyridine pulse, 20 s exposure, and 30 s purge. A total of 625 cycles was applied, corresponding to a ZnF₂ growth rate of ~0.8 Å per cycle on a reference Si substrate.

Material characterization

The surface morphology were examined by field-emission scanning electron microscopy (FE-SEM) using JSM-7800F and IT-800, JEOL. Elemental distributions were analyzed using an energy-dispersive X-ray spectroscopy (EDX) system (AZtec®, Oxford Instruments) coupled to the SEM or STEM. X-ray diffraction (XRD) patterns were recorded on an Empyrean diffractometer (PANalytical) equipped with Cu Kα radiation ($\lambda = 1.5418 \text{ Å}$).

X-ray photoelectron spectroscopy (XPS) measurements were carried out using an EnviroESCA spectrometer from Specs GmbH – Phoibos 150 analyzer suitable for near-ambient pressure, 1D Delay-Line Detector, monochromatic Al Kα radiation at 1486.6 eV, incident angle 55°. Cross section profiling was done with an Ionoptika GCIB 10S (10 kV, Ar⁺ ion sputtering, 3 × 3 mm²). The samples were prepared in a glovebox and transferred under argon into the measurement system. To prevent from unwanted surface charging, the sample's back side and its surface were connected to the ground sample carrier by using conductive carbon tape and a molybdenum clamp. The measurements were conducted at room temperature and under high vacuum ($p \leq 7 \times 10^{-7} \text{ mbar}$).

Survey spectra were measured utilizing a pass energy of 100 eV (10 scans), high-resolution spectra for all elemental regions were obtained at a pass energy of 30 eV (30 scans). The C 1s signal was used as binding energy reference (284.8 eV), for the spectra of the depth profile without carbon, the Zn^{2+} 2p_{3/2} (1022.5 eV) signal was used as reference. The calibration of the energy scale was performed using the Au 4f_{7/2} (83.96 eV) and Ag 3d_{5/2} (368.21 eV) photoelectron peaks.

The XPS data was processed with XPSPEAK41 by applying the following peak fitting settings: A Shirley background and a Gaussian-Lorentzian line shape GL(20%) were utilized to fit all signals; a doublet separation energy for the Zn 2p region of 23.2 eV was applied. For semi-quantitative investigations relative sensitivity factors from XPSPEAK41 were used.

Cross-sectional TEM specimens were prepared by cryo-FIB milling. The detailed cryo-FIB specimen preparation and cryogenic transfer procedures are described elsewhere (S1) The relative specimen thickness was evaluated from the simultaneously acquired low-loss EELS spectra using the log-ratio method ($t/\lambda = \ln(I_t/I_0)$), yielding $t/\lambda \approx 0.1\text{--}0.5$ across the reaction layer, confirming that the specimen was sufficiently thin for reliable core-loss EELS analysis. STEM imaging and EELS analysis were performed on a probe-Cs-corrected transmission electron microscope (Titan G2, Thermo Fisher) operated at 300 kV. Cross-sectional ADF-STEM images were acquired using a Gatan ADF detector (Model 806) at a camera length of 60 mm. The convergence semi-angle of the STEM probe was 20 mrad.

EELS data were acquired in dual-EELS mode using a Gatan GIF Quantum ERS spectrometer with an energy dispersion of 0.5 eV/channel and a 5 mm entrance aperture, corresponding to a collection semi-angle of @@ mrad. Line profiles of 100 pixels were recorded across the reaction layer with a pixel spacing of ~5.5 nm and a dwell time of 0.4 s per pixel for the high-loss region. The F K-edge (~685 eV), Zn L-edge (~1020 eV), and Na K-edge (~1072 eV) were used to determine the spatial distribution of each element.

The Cryo-STEM experiments were conducted to minimize beam-induced degradation of the Na-containing reaction layer. Specimens were transferred from an Ar-filled glovebox to the TEM at room temperature using a Melbuild Cryo-DT Atmos vacuum transfer holder. Once loaded into the TEM, the specimen was cooled to $-170\text{ }^{\circ}\text{C}$ for imaging and EELS analysis.

Na||Al and Na|| ZnF₂/Al cell assembly and electrochemical measurements

Half cells were assembled in 2032-type coin cells using bare Al or ZnF₂/Al as the working electrode and Na metal discs as the counter/reference electrode (denoted Na||Al and Na||ZnF₂/Al, respectively). The electrolyte was 1.0 M NaPF₆ dissolved in diglyme, and a Celgard 2500 membrane was used as the separator. All cells were first discharged to 0.01 V vs Na⁺/Na to activate the ZnF₂ layer, and then cycled five times between 0.01 and 1.0 V at 0.01 mA cm⁻² to stabilize the SEI before Na deposition tests. Subsequently, galvanostatic Na plating/stripping was performed at constant current densities of 0.5 and 2.0 mA cm⁻² to evaluate nucleation overpotentials, Coulombic efficiencies, and morphological evolution.

Symmetric cell assembly and electrochemical measurements

For symmetric-cell tests, Na/ZnF₂/Al discs with a pre-deposited Na areal capacity of 2.0 mAh cm⁻² were used as both electrodes. The electrolyte and separator were the same as in the half-cell tests (1.0 M NaPF₆ in diglyme and Celgard 2500). Symmetric cells were cycled galvanostatically at 1.0 mA cm⁻² with a cut-off capacity of 1.0 mAh cm⁻² per half-cycle to assess long-term Na plating/stripping stability and voltage hysteresis.

Na||NVP full cell assembly and electrochemical measurements

 Na₃V₂(PO₄)₃/C material was synthesized by ball-milling and carbothermal reduction method, as described elsewhere (S2) Poly(acrylic acid) (PAA) and d-(+)-glucose act as carbon sources and reducing agents. In detail, the starting materials from Sigma Aldrich: Na₂CO₃ (3.675 mmol), NH₄VO₃ (4.90 mmol), and NH₄H₂PO₄ (7.35 mmol) with a molar ratio of

(Na:V:P = (3:2:3), were added to 5 mL of ethanol together with PVA (3.5mg) and d-(+)-glucose (101 mg). The mixture was ball-milled using PULVERISETTE 7 with 60 ZrO₂ spheres of 5 mm (per 20 ml cup) for 8 h at 350 rpm. Ball-milling was interrupted every 5 min and the system was let to rest for 7 min, in order to avoid the heating of the solution. The solvent was evaporated under N₂ atmosphere at 60 °C for 12 h and the remaining powder was ground for 1 h, followed by an annealing in two steps in Ar:H₂ atmosphere (5% H₂): the powder was first heated at 350 °C for 5 h with a heating rate of 5 °C min⁻¹ and then was kept at 800 °C for 8 h with a heating rate of 3 °C min⁻¹. Finally, the obtained black powder was ground with mortar and pestle for 1 h.

NVP was used as the positive electrode material. NVP powder, Super P conductive carbon, and polyacrylic acid (PAA) binder were mixed in deionized water at a mass ratio of 8:1:1 to form a homogeneous slurry. The slurry was cast onto Al foil and dried at 80 °C overnight, yielding a cathode with an active-material loading of ~11.5 mg cm⁻². The dried composite electrodes were punched into 14 mm-diameter disks and used as cathodes. For anode-less cells, Na was pre-deposited onto bare Al or ZnF₂/Al (Na/Al and Na/ZnF₂/Al) to an areal capacity of 2.0 mAh cm⁻², and these Na-plated foils were paired with NVP cathodes. For anode-free cells, bare Al or ZnF₂/Al foils without pre-deposited Na were used directly as anode-side current collectors and coupled with NVP cathodes. In all full cells, the electrolyte was 1.0 M NaPF₆ in diglyme, and a stack of Celgard 2500 plus a glass-fiber membrane was used as the separator. Galvanostatic cycling was performed on a battery cycler (Wonatech) between 2.8 and 3.9 V at room temperature at various C-rates as specified in the main text.

Computational calculations

Density functional theory (DFT) calculations were performed using the Vienna *ab initio* Simulation Package (VASP) (S3) to compare the sodium adsorption preferences on the ZnF₂, NaF, and NaZn₁₃ surfaces. The Perdew–Burke–Ernzerhof (PBE) form of the generalized gradient approximation (GGA) (S4) was employed for the exchange–correlation functional,

and the k-point sampling and plane-wave cut-off energy were determined to satisfy convergence criteria of free energy within 10 meV/atom and pressure within 10 kBar for all structures. The surface energy (γ) was calculated by the following equation:

$$\gamma = \frac{E_{\text{slab}} - nE_{\text{bulk}} - \Delta N_x \mu_x}{2A}$$

where the E_{slab} is the total energy of the relaxed slab model, nE_{bulk} represents the energy of n formula units of the corresponding bulk structure, ΔN_x is the number of excess (or deficient) atoms of species x relative to the stoichiometric bulk, μ_x is the chemical potential of species x , and A is the surface area. The factor of 2 accounts for the two equivalent surfaces present in the symmetric slab model. The adsorption energy (E_{ads}) of Na was calculated by following equation:

$$E_{\text{ads}} = E_{\text{Na+slab}} - E_{\text{Na}} - E_{\text{slab}}$$

where $E_{\text{Na+slab}}$ represents the total energy of Na-adsorbed slab structure and E_{Na} is the chemical potential of Na bcc metal phase.

2. Supporting Figures

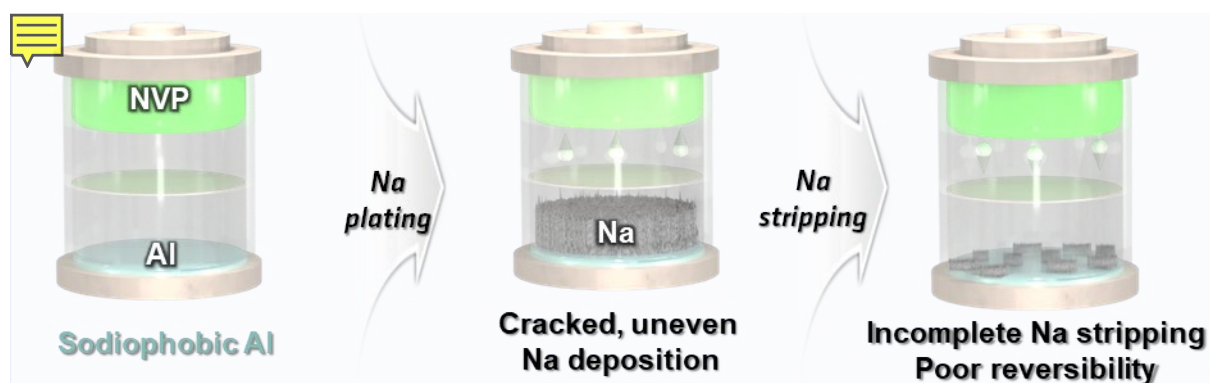


Figure S1. Schematic illustration of an anode-free sodium battery using bare Al as the anode-side current collector. During charging, Na^+ supplied from the NVP cathode is plated onto the Al surface, forming non-uniform Na deposits. Upon discharge, incomplete stripping of the deposited Na leads to residual Na and poor reversibility.

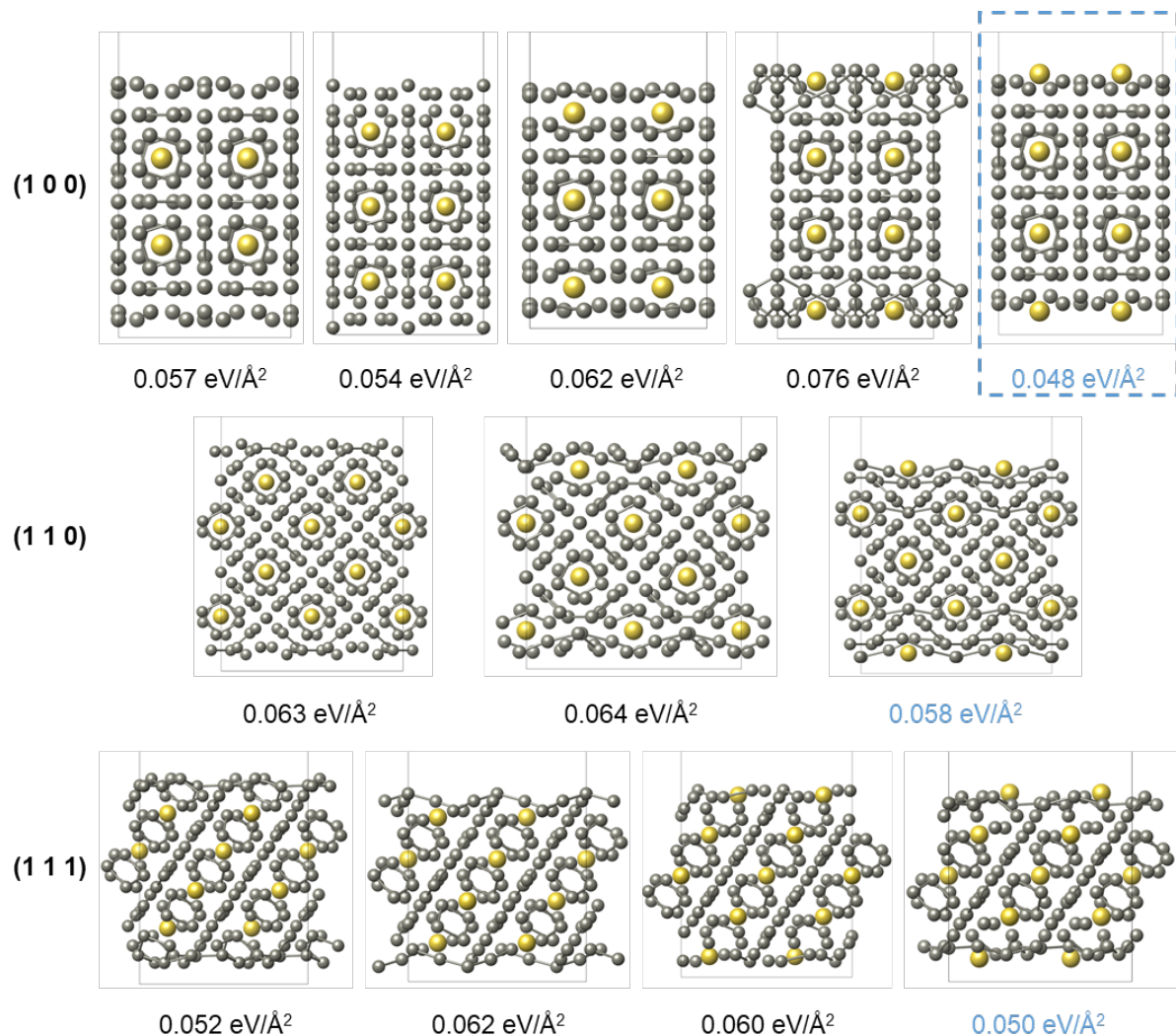


Figure S2. Identification of the most stable surface among various NaZn_{13} facets under Zn-rich conditions. DFT-optimized surface structures of NaZn_{13} for the (100), (110), and (111) facets with different outermost layer terminations, together with their corresponding surface energies ($\text{eV}/\text{\AA}^2$). The lowest-energy configurations (highlighted for the (100), (110), and (111) planes) are used as the representative surfaces for subsequent Na adsorption calculations.

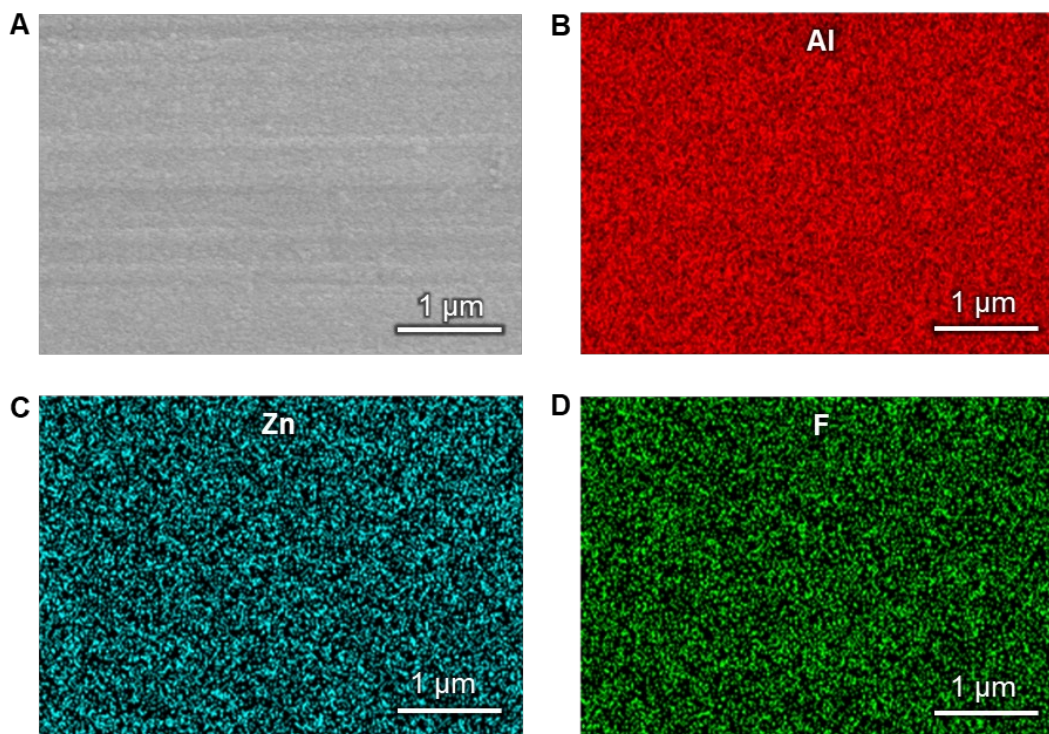


Figure S3. Surface morphology and elemental distribution of the ALD-ZnF₂ coating on Al. **A)** SEM image of the ZnF₂/Al surface. **B–D)** Corresponding EDX elemental maps of Al, Zn, and F, respectively.

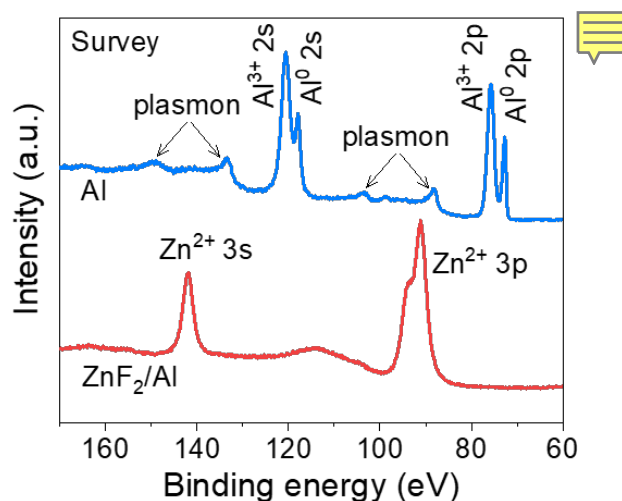


Figure S4. XPS survey spectra of bare Al and ZnF₂/Al. The bare Al sample shows prominent Al 2s and Al 2p peaks of oxidic and metallic species, and additionally plasmon peaks from the metallic Al, whereas the ZnF₂/Al sample exhibits Zn 3s and Zn 3p signals and no detectable Al peaks.

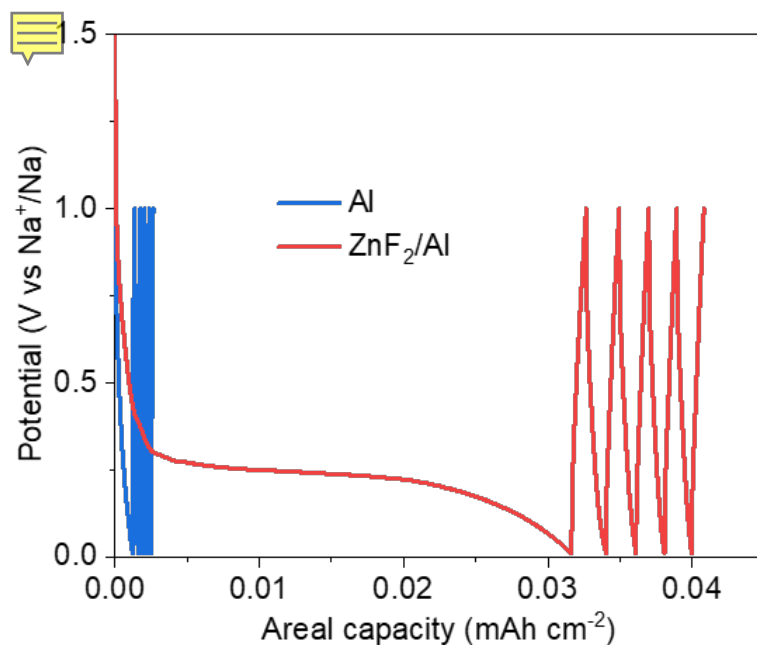


Figure S5. Potential profiles of the cells in the initial sodiation process. The cells were first discharged to 0.01 V and then cycled five times between 0.01 and 1.0 V to form and stabilize the SEI.

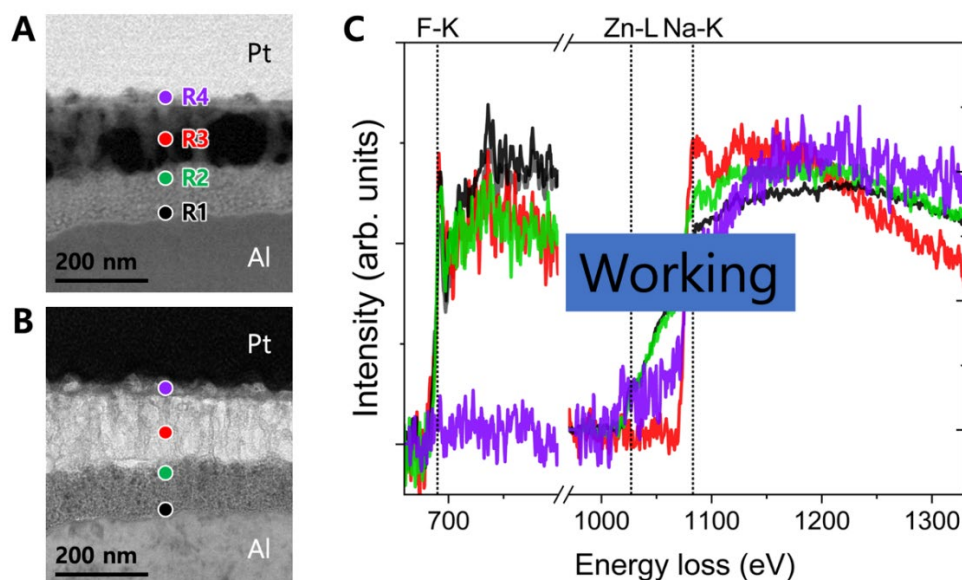


Figure S6. Cross-sectional STEM/TEM images and representative EELS spectra of the sodiated ZnF_2/Al interphase. **A)** ADF-STEM image acquired during EELS spectrum imaging, with labeled extraction points (R1–R4). **B)** BF-TEM image of the same region, showing complementary contrast to (A). **C)** EELS spectra from R1–R4 showing the F K-edge (~685 eV), Zn L-edge (~1020 eV), and Na K-edge (~1072 eV). The relative edge intensities suggest a layered phase sequence from the Al substrate outward: (R1) unreacted ZnF_2 , (R2) NaF , (R3) partially converted $\text{ZnF}_2/\text{NaF}/\text{Zn}$, and (R4) NaZn_{13} .

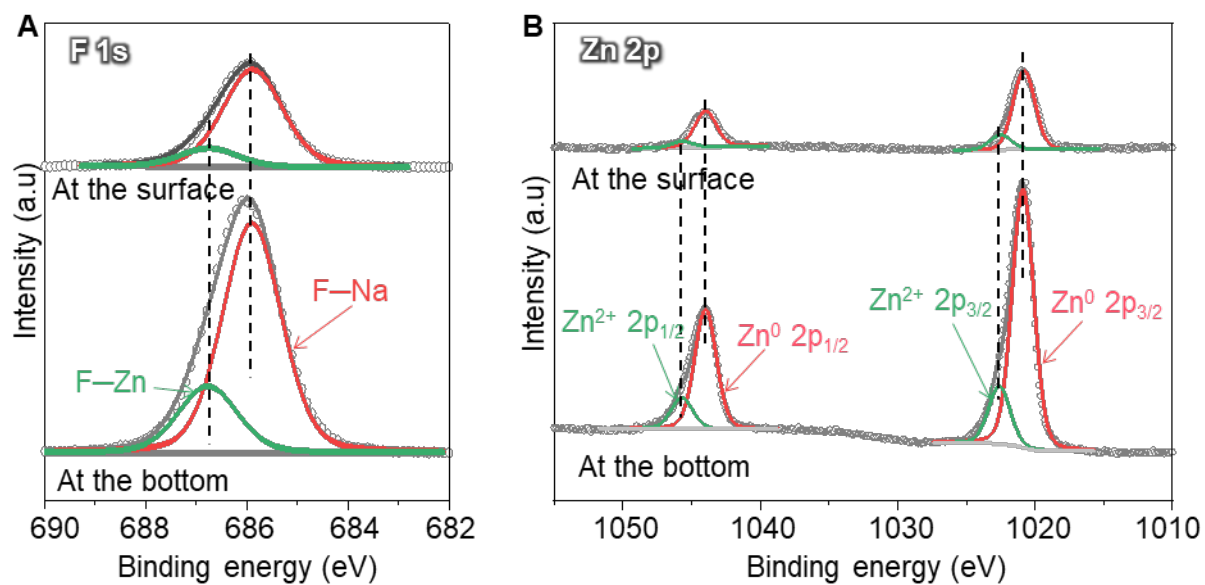


Figure S7. High-resolution XPS spectra of sodiated ZnF₂/Al obtained at the surface and bottom of the NaF/NaZn₁₃ layer. A) Fitted F 1s spectra. B) Fitted Zn 2p spectra.

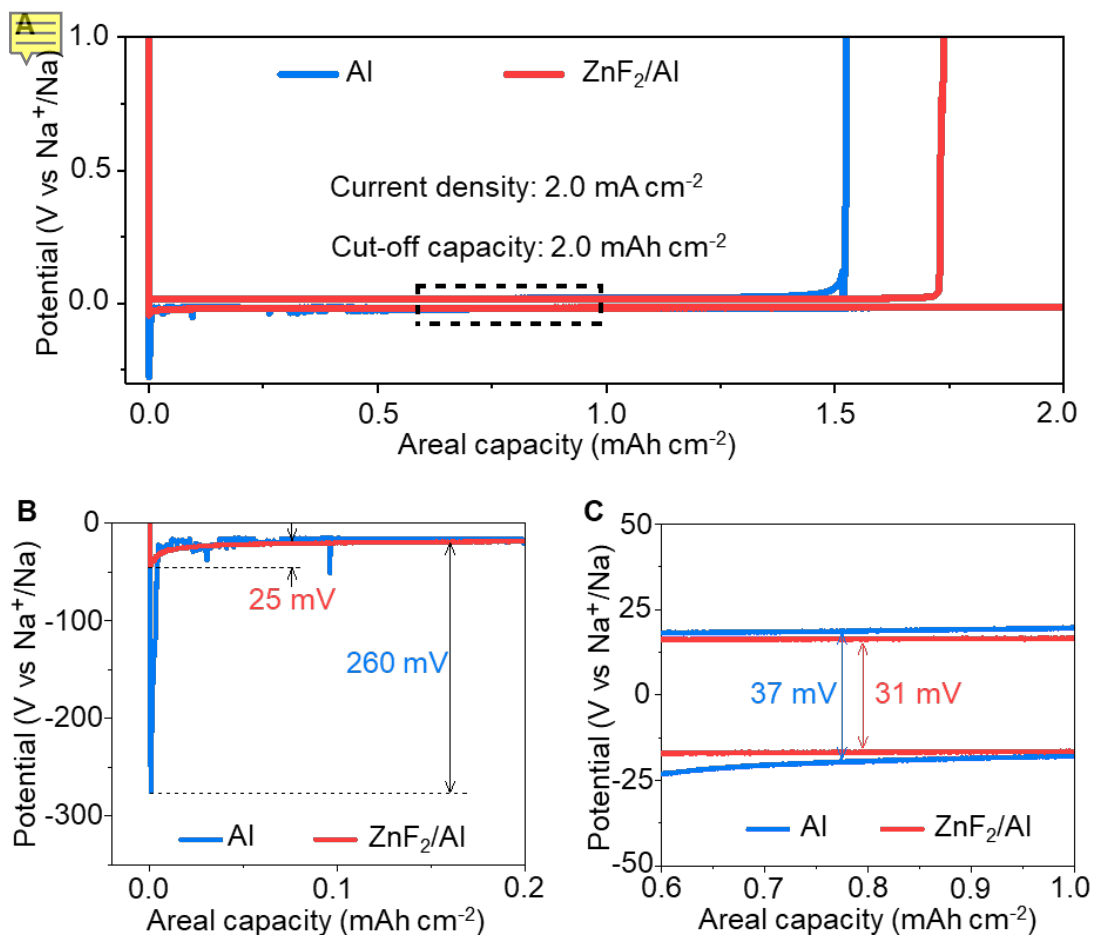


Figure S8. Na plating/stripping behavior on bare Al and ZnF_2/Al at high current density. **A)** Voltage profiles of Na plating and stripping on bare Al and ZnF_2/Al at 2.0 mA cm^{-2} with a cut-off capacity of 2.0 mAh cm^{-2} . **B)** Enlarged views of the initial nucleation region and the subsequent stripping plateau. **C)** Enlarged voltage–capacity profiles in the boxed region of **(A)**, comparing the steady-state plating/stripping hysteresis of Na on bare Al and ZnF_2/Al at 2.0 mA cm^{-2} .

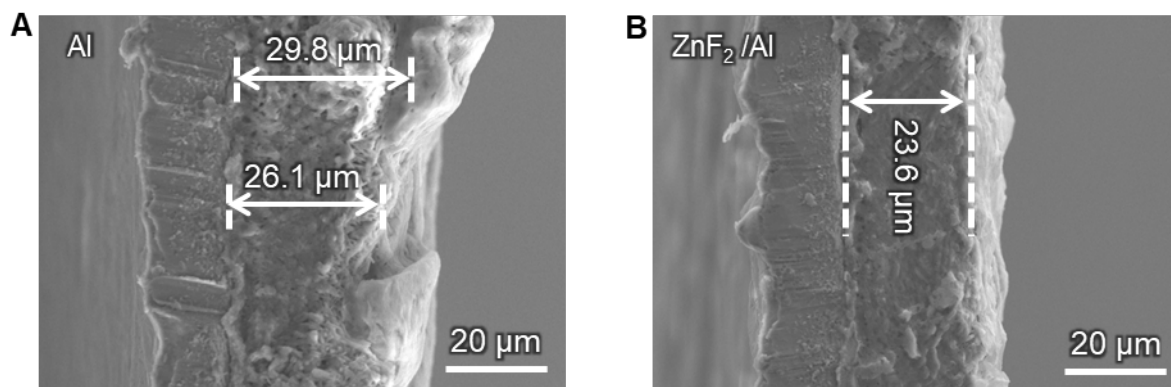


Figure S9. Cross-sectional SEM analysis of Na layer thickness on different current collectors. A) Na deposited on bare Al to 2 mAh cm⁻², showing a thick and highly non-uniform Na layer with local thicknesses of ~26–30 μm. **B)** Na deposited on ZnF₂/Al at the same areal capacity.

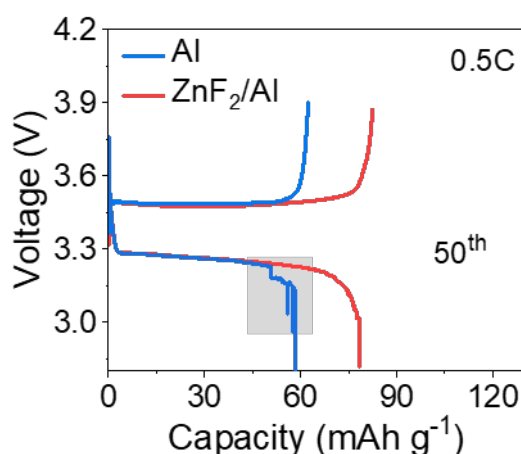


Figure S10. Charge–discharge profiles of AFSB cells with the NVP cathode at 0.5 C at the 50th cycle. The grey region highlights irregular voltage fluctuations of the Al||NVP cell.

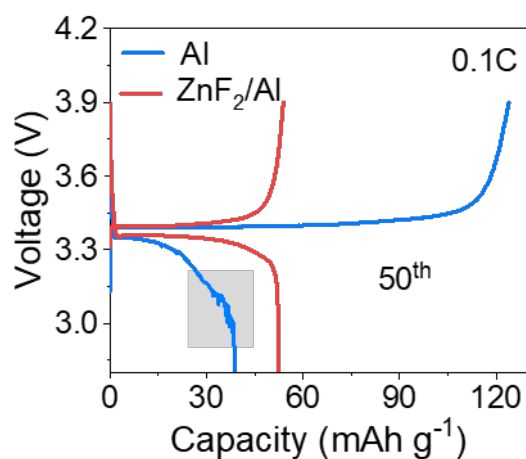


Figure S11. Charge–discharge profiles of AFSB cells with the NVP cathode at 0.1 C at the 50th cycle. The grey region highlights irregular voltage fluctuations of the Al||NVP cell.

3. Supporting Table

No.	Element	Binding energy (eV)	Reference
1	Zn ²⁺ 2p _{3/2}	1022.5	S5
2	Zn ²⁺ 2p _{1/2}	1045.6	S5
3	Zn ⁰ 2p _{3/2}	1021.6	S5
4	Zn ⁰ 2p _{3/2}	1044	S5
5	F–Zn	685.2	S6
6	F–Na	684.1	S7
7	Na ⁺ 1s	1071.9	S8
8	Na ⁰ 1s	1071.4	

4. References

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